

What survives refinement: results on Newton's vanishing sagitta and the halving of lattice gauge fields

Abstract. We collect the theorems obtained in this repository on two refinement limits. In Newton's comparison of free motion with motion under a constant force, the action of a cell of duration τ is spent additively by cuts, and a record of the comparison made by marks at those cuts reaches exactly the spent action divided by the mark floor κ . The best verdict on the comparison is continuous in the force if and only if $\kappa > 0$, which identifies a positive floor with Leibniz's law of continuity read on records. For lattice gauge theories refined by halving, one directional halving factors into series moves and a mid-plane integral; for $SU(2)$ in $1 + 2$ dimensions one halving step is compared with a covariant Gaussian reference, uniformly in the size of the plane, with responses controlled through third order and large-field clusters shown to be jointly rare. The one-loop coefficients transfer to $SU(N)$ with the factor $N/2$. Each statement lists its hypotheses; complete proofs are in the notes cited after each proof sketch. Planck's constant $\hbar$ and all dimensional constants are kept explicit.

1. Newton's comparison and the cut measure

Definition 1.1 (Galileo cell). A body of mass m moves on $[0, \tau]$ either freely or under a constant force F , from the same position and velocity. The two positions differ by $P(t) = Ft^2/(2m)$. The *cell action* is

$$K_\tau = \frac{F^2 \tau^3}{24m}.$$

Theorem 1.2 (cut measure). Cutting the cell at the fraction $s \in (0, 1)$ of its duration spends the action $3s(1-s)K_\tau$, that is $K_\tau = 3s(1-s)K_\tau + K_{s\tau} + K_{(1-s)\tau}$. For any finite set of cuts with pieces τ_i ,

$$K_\tau = \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}_{\text{cut}}^2 + \sum_i K_{\tau_i},$$

where ϕ_{cut} is the Schauder hat inserted by the cut. The spent action tends to K_τ if and only if $\max_i \tau_i \rightarrow 0$.

Proof (sketch). The hat of a cut at $s\tau$ has height $h = Fs(1-s)\tau^2/(2m)$ and energy $\frac{m}{2}h^2(\frac{1}{s\tau} + \frac{1}{(1-s)\tau}) = 3s(1-s)K_\tau$; later hats are orthogonal to earlier ones in the Cameron–Martin space H_0^1 , and $\sum_i K_{\tau_i} \leq K_\tau \max_i \tau_i^2/\tau^2$. Cut-measure note, Theorems 1–2 and Proposition 4. $\square$

Corollary 1.3. Halving every piece n times spends $(1-4^{-n})K_\tau$. Halving only the remaining piece, day after day, spends $\frac{6}{7}(1-8^{-n})K_\tau \rightarrow \frac{6}{7}K_\tau$; two quarter-cuts a day spend $\frac{27}{28}K_\tau$.

2. Records and the floor

Definition 2.1 (mark model). A protocol places marks at times $t_0 < t_1 < \dots < t_N$ fixed in advance. Mark j returns the position with an independent centred Gaussian error of standard deviation δ_j and delivers an independent centred Gaussian impulse of standard deviation Δ_j , with

$$\delta_j \Delta_j \geq \kappa \geq 0.$$

The two hypotheses are read by tests invariant under the unknown initial position and velocity. The optimal such test errs with probability $\Phi(-d/2)$, where d is the statistical distance of the two hypotheses.

Theorem 2.2 (the floor). For every protocol of Definition 2.1, $d^2 \leq K_\tau/\kappa$, independently of the number and times of the marks; the constant is sharp. Deciding at error ϵ therefore requires $F^2 \tau^3 \geq 96 z_{1-\epsilon}^2 m \kappa$, with $z_{1-\epsilon} = \Phi^{-1}(1-\epsilon)$.

Proof (sketch). Write the test weights as a piecewise-linear T vanishing outside (t_0, t_N) ; the signal is $\frac{F}{m} \int T$ and the noise is at least $\frac{2\kappa}{m} \int T'^2$ by the arithmetic-geometric mean; the sharp inequality $(\int T)^2 \leq \frac{\tau^3}{12} \int T'^2$ on H_0^1 closes the bound. Planck paper, Theorem 2. $\square$

Theorem 2.3 (records spend the cut measure). Assume marks with $\delta\Delta = \kappa > 0$ are available at every $\delta > 0$. For marks at $t_0 < \dots < t_N$, with $\tau = t_N - t_0$ and pieces τ_i ,

$$\sup d^2 = \frac{K_\tau - \sum_i K_{\tau_i}}{\kappa},$$

approached with sharp end marks and balanced interior marks. A single cut at the fraction s carries $3s(1-s)K_\tau/\kappa$, and adding a mark inside a piece adds exactly the share of that cut.

Proof (sketch). The maximum of $(\int T)^2 / \int T'^2$ over piecewise-linear T on the grid is attained by the Galerkin solution of $-T'' = 1$, which is exact at the nodes; its integral is the trapezoid rule for $\frac{1}{2}(w - t_0)(t_N - w)$, whose error is $\sum_i \tau_i^3/12$. Cut-measure note, Proposition 7. $\square$

Theorem 2.4 (any instrument, bounded aperture). Let instruments with any Kraus operators, any apparatus memory and arbitrary adaptivity act at times in $(0, \tau]$, identical under the two hypotheses, followed by any final measurement, and let every conditional state keep the body within position and momentum spreads L and P . If the record decides at error ϵ , then

$$\frac{F\tau L}{\hbar} + \frac{F\tau^2 P}{2m\hbar} \geq 1 - 2\epsilon.$$

Probabilistic note, Theorem 2 (hybrid argument over displacement operators).

Theorem 2.5 (recoil). For every protocol of momentum-transfer marks with pointers read, with probes in arbitrary states, deciding at error ϵ requires $s \sum_j \Delta_j \geq 8\hbar \arcsin(1 - 2\epsilon)$, with $s = F\tau^2/(2m)$ the sagitta and Δ_j the spread of the impulse delivered by mark j . Recoil note, Theorem R, with the constant of the path-length note.

3. Continuity of the verdict

Theorem 3.1 (continuity). In the mark model, the best error with which the comparison is decided is

$$P_*(F) = \Phi\left(-\frac{1}{2}\sqrt{K_\tau/\kappa}\right) \quad (\kappa > 0), \quad P_*(F) = 0 \quad (F \neq 0), \quad P_*(0) = \frac{1}{2} \quad (\kappa = 0).$$

Hence P_* is continuous at $F = 0$ if and only if $\kappa > 0$. The same holds for rest against uniform motion with $mv^2\tau/(8\kappa)$ in place of $K_\tau/(4\kappa)$. For a static taper, read without back-action, P_* jumps at zero for every $\kappa \geq 0$.

Proof (sketch). Theorem 2.2 and its sharpness give the value for $\kappa > 0$; for $\kappa = 0$ three marks with zero impulse spread and vanishing resolution decide any $F \neq 0$. Continuity note, Proposition L. $\square$

Corollary 3.2 (topologies). The supremal statistical distance between the recorded motions of a cell under forces F, F' is $d(F, F')^2 = (F - F')^2 \tau^3 / (24m\kappa)$ for $\kappa > 0$ and infinite off the diagonal for $\kappa = 0$. Thus $\kappa > 0$ if and only if records and the geometry of the motions induce the same topology on the constant-force family.

Corollary 3.3 (known preparations). Under the hypotheses of Theorem 2.4, $P_*(F) \geq \frac{1}{2}(1 - F\tau L/\hbar - F\tau^2 P/(2m\hbar))$, so at bounded aperture the best verdict is continuous at $F = 0$ whenever $\hbar > 0$.

Remark 3.4 (Leibniz, 1687). Leibniz's law of continuity requires that when two cases approach, their outcomes approach (*Nouvelles de la République des Lettres*, July 1687; Gerhardt III, 52). Theorem 3.1 shows that, with outcomes measured by how well they can be told apart, the law holds for Newton's comparison exactly when $\kappa > 0$.

Proposition 3.5 (the floor as a path-measure threshold). For two Gaussian bridges of variance rate $\hbar/m$ around the chord and the parabola, the optimal equal-prior test errs

with probability $\Phi(-\sqrt{K_\tau/(2\hbar)})$, and decides at error ϵ iff $K_\tau \geq 2z_{1-\epsilon}^2\hbar$. Cut-measure note, Proposition 6.

Proposition 3.6 (one constant). Brownian free motion with continuous stationary independent increments and isotropic centred noise, mass-only laws and independent centre-of-mass composition for all positive masses force $mD = \kappa$ with a single constant κ for all bodies. Stochastic-route note, S1–S3.

4. Halving lattice gauge fields

Consider a heat-kernel lattice gauge theory in D dimensions with compact gauge group G , lattice spacing a and heat time $t = \lambda_D a$ per plaquette.

Theorem 4.1 (series/parallel factorization). One halving of a lattice direction is exactly the product of series moves, heat-kernel convolutions that close in every dimension, and a parallel factor Ψ , a $(D-1)$ -dimensional gauge theory on the mid-plane whose edge laws are Brownian bridges. It has $\binom{D-1}{2}$ transverse planes. Series/parallel note, Proposition 1.

Theorem 4.2 (free defect). For the free field the defect $\mathcal{D}$ of one halving is an exact nonpositive quadratic form of relative size at most $a^2 K^2/16 + (Ga)^2/8$; for a cut at the fraction s with trapezoid weights the bound scales exactly by $4s(1-s)$. Series/parallel note, Proposition 2 and Corollary 2_s.

Theorem 4.3 ($U(1)$ in three dimensions). One unperturbed halving step equals the free step up to an extensive error of density $e^{-\pi^2/(8\lambda_3 a)}$ on the small-field set, for every cut fraction. The whole measure lies within total variation $2(L/a)^3 e^{-\pi^2/(6\lambda_3 a)}/(1 - e^{-\pi^2/(2\lambda_3 a)})$ of its monopole-free part. Series/parallel note, Theorem 5 and Corollary 5_s; monopole note, Theorem 2.

Theorem 4.4 (bridge midpoints and the centre). For a compact connected simply connected group G , the character moments of a Brownian-bridge point at any cut fraction, and at several cuts, are exact sums over weights and coroot images. At the reduced fraction $s = p/q$ the image terms are central iff $(1-s)H \in P^\vee$; for $SU(N)$ they reach the subgroup $\mathbb{Z}_{\gcd(q,N)}$ of the centre, so the centre of $SU(3)$ is reached only when $3 \mid q$. Centre note, Theorems 1 and 1'.

5. The $SU(2)$ mid-plane in $1+2$ dimensions

Setting 5.1. Halve the time direction of the heat-kernel $SU(2)$ theory in $1+2$ dimensions with coarse heat time $t = \lambda_3 a \leq t_0$, on a periodic mid-plane with sides at least three. In the mid-vertex gauge write $m_e = m_{*e} e^{\xi_e}$. Fix $\delta \in (0, 1/10)$ and put $\varepsilon = t^{1/2-\delta}$, $\alpha' = \frac{1}{2} - 3\delta$. The boundary data are small: face logarithms $|x_p|, |y_p| \leq \varepsilon$ in the two layers and cut data $|X_e| \leq \varepsilon$. The measure carries a convex radial barrier vanishing for $|\xi_e| \leq 2\varepsilon$ and diverging at the chart radius; the smallness conditions on t and the chart are those of the small-field note, (65), (83), (98) and §22.3. Let C_{m_*} be the incidence operator with adjoint transports of the midpoint connection, $P = (8 + C_{m_*} C_{m_*}^*)^{-1}$ and $H = 4 + \frac{1}{2} C_{m_*}^* C_{m_*}$.

Definition 5.2 (covariant reference).

$$\mathcal{D}^{\text{cov}} = \frac{1}{2t} \langle x + y, P(x + y) \rangle - \frac{\|x\|^2 + \|y\|^2}{8t} + \frac{1}{2} \log \frac{\det H}{\det H_1}.$$

Theorem 5.3 (one step). Uniformly in the size of the plane,

$$|\mathcal{D}_s - \mathcal{D}^{\text{cov}}| \leq C t^{\alpha'} \left[\sum_p \left(1 + \frac{|x_p|^2 + |y_p|^2}{t} \right) + \sum_e \left(1 + \frac{|X_e|^2}{t} \right) \right],$$

where $\mathcal{D}_s$ is the normalized log-ratio of the barriered mid-plane integral, including Haar and heat-kernel amplitudes, and C depends only on local constants of the compact chart.

Proof (sketch). Classical comparison at the saddle, a local bound on the saddle determinant, uniform saddle-centred moments by integration by parts with diagonal dominance, the Laplace

remainder by interpolation, and a Gaussian rectangle inequality for the barrier. Small-field note, (58) and (62). $\square$

Theorem 5.4 (responses through third order). Let $r = \mathcal{D}_s - \mathcal{D}^{\text{cov}}$, and let ∂_a change the boundary link a at velocity ε along an admissible coordinate box. With $\gamma = \frac{1}{2} \log(10/7)$ and the tree length ℓ of the labels, for $n = 1, 2, 3$,

$$\sup_{a_1} \sum_{a_2, \dots, a_n} e^{\gamma \ell(a_1, \dots, a_n)} |\partial_{a_1} \cdots \partial_{a_n} r| \leq C_n t^{\alpha'},$$

including adjacent and coincident links, uniformly in the plane and in the barrier approximation.

Proof (sketch). Helffer–Sjöstrand representations of the covariances, a Feynman–Kac representation of the resolvent with pointwise block bounds, the differentiated resolvent equation, the barrier removed one edge at a time, gradient extensions of the scores, and the cubic vanishing of the local energy difference $S - V_2$. Small-field note, (89), (93), (119). $\square$

Theorem 5.5 (large-field clusters). With $p = 2(7/16)^{k_t}$ and $k_t = \lfloor 3t^{-2\delta}/8 \rfloor - 1$, for every set H of edges and every choice of the barrier edges,

$$\nu(|\xi_e| > \eta \text{ for all } e \in H) \leq p^{|H|/16},$$

and for disjoint sets H_1, H_2 at graph distance d the insertion products $F_H = \prod_{e \in H} (e^{-w(\xi_e)} - 1)$ satisfy

$$|E F_{H_1 \cup H_2} - E F_{H_1} E F_{H_2}| \leq C p^{(|H_1| + |H_2|)/64} (5/12)^d.$$

Proof (sketch). Exponential moments of linear functionals under the uniformly convex measure, stable under linear tilts, and a Chernoff bound; the correlation by the Helffer–Sjöstrand covariance decay. Small-field note, (111) and (115). $\square$

6. From $SU(2)$ to $SU(N)$

Theorem 6.1 (one-loop coefficients). With the metric $\langle X, Y \rangle = -2 \text{tr} XY$, the relative-order- t coupling coefficients of one halving for $SU(N)$ are those of $SU(2)$ multiplied by $N/2$:

$$\delta_t^{12} = \delta_t^{13} = \frac{Nt}{48} \int_{\mathcal{B}} \frac{q}{R}, \quad \delta_t^{23} = \frac{N}{2} t \left(-\frac{1}{12} + 2\mathcal{I} \right),$$

with the group-independent integrals of the order- t note. For $SU(3)$ the cut couplings shift by $\frac{t}{16} \int q/R > 0$ and the transverse coupling by $t(-\frac{1}{8} + 3\mathcal{I}) < 0$.

Proof (sketch). Every group factor enters through the Killing form, $\text{tr}_{\text{adj}}(\text{ad } X)^2 = -N|X|^2$; around a Cartan background the plaquette Hessian splits into root spaces, each an isometrically embedded $SU(2)$ of charge $\alpha(Y)$, and $\sum_{\alpha > 0} \alpha(Y)^2 = \frac{N}{2}|Y|^2$. Order- t note, §7b. $\square$

Proposition 6.2 (twisted $SU(3)$ sector). For the clock and shift matrices $P, Q \in SU(3)$, $PQ = \omega QP$ with $\omega = e^{2\pi i/3}$, the adjoint action has the eight joint phases $(2\pi a/3, -2\pi b/3)$, $(a, b) \in \mathbb{Z}_3^2 \setminus \{0\}$. The twisted flat sector therefore has no adjoint zero mode, and its mid-plane Laplacian is bounded below by $2 - 2 \cos(2\pi/(3n_{\text{max}}))$. Small-field note, §16.

Proposition 6.3 (bridge tails for any compact group). A bi-invariant metric has nonnegative Ricci curvature, so the Li–Yau bounds give, for the bridge midpoint of duration $t \leq 1$ and every $\epsilon' \in (0, 1)$, $\beta\{d(m, m_*) \geq r\} \leq C_{\epsilon'} t^{-n/2} \exp\{-[4(r - |X|/2)^2/(2 + \epsilon') - |X|^2/(2 - \epsilon')]/t\}$, with $n = \dim G$. Small-field note, §16.

7. Consequence for STATE

This note is the formal compendium of the results above, written for readers who want the statements with their hypotheses in one place; the cited notes hold the proofs and remain the record. It is linked from the site’s index and tutorial.